

# AI Face Mask Detection System

Project Report — Real-Time Computer Vision and Deep Learning Application

## 1. Introduction

The AI Face Mask Detection System is a real-time computer vision application developed to identify whether a person is wearing a face mask. The system uses OpenCV to detect faces from a webcam feed and a trained Convolutional Neural Network (CNN) to classify each detected face as With Mask or Without Mask. A Flask-based web dashboard presents the live camera feed, prediction status, confidence score, and number of detected faces.

## 2. Abstract

This project demonstrates the complete development of an AI-powered real-time monitoring application. Face images are organized into masked and unmasked categories, preprocessed, and used to train a CNN model. The trained model is saved as **face\_mask\_detector.keras** and integrated with OpenCV for live prediction. During operation, Haar Cascade face detection identifies faces, the CNN predicts mask usage, and the result is displayed with a confidence percentage. The Flask application also provides detection status and history through web interfaces and APIs. The trained model achieved a test accuracy of **96.30%**.

## 3. Tools Used

| Tool / Technology       | Purpose                                              |
|-------------------------|------------------------------------------------------|
| Python                  | Core programming and application logic               |
| TensorFlow / Keras      | CNN model training and prediction                    |
| OpenCV                  | Face detection, webcam capture, and image processing |
| NumPy                   | Numerical and image-array operations                 |
| Pillow                  | Image handling and preprocessing support             |
| Flask                   | Web application and API development                  |
| Matplotlib              | Model evaluation visualizations                      |
| Scikit-learn            | Evaluation metrics and analysis                      |
| HTML / CSS / JavaScript | Interactive web dashboard                            |
| Git / GitHub            | Version control and repository management            |

## 4. Steps Involved in Building the Project

- **Dataset preparation:** Organized images into *with\_mask* and *without\_mask* categories.
- **Image preprocessing:** Resized and normalized images to create consistent CNN input.
- **CNN training:** Developed and trained a deep learning model to classify mask usage.
- **Model evaluation:** Tested the trained model and generated accuracy and confusion-matrix reports. The reported test accuracy was 96.30%.
- **Face detection:** Integrated OpenCV Haar Cascade to locate faces in live webcam frames.
- **Real-time prediction:** Each detected face is resized, converted to the required format, normalized, and passed to the CNN.
- **Flask web application:** Connected the AI pipeline to a browser-based dashboard with live video streaming and detection APIs.
- **Dashboard and history:** Added confidence, face count, status indicators, and a detection history interface.
- **Testing and version control:** Verified Python syntax, tested the application locally, documented the project, and maintained the source code using Git and GitHub.

## 5. Conclusion

The AI Face Mask Detection System successfully combines deep learning, computer vision, and web development into a practical real-time application. The system can detect faces through a webcam, classify mask usage, display prediction confidence, and maintain detection information through a Flask dashboard. The 96.30% test accuracy demonstrates the effectiveness of the trained model on the evaluated test data. The completed project provides a strong practical example of applying CNN-based image classification and OpenCV-based face detection to a real-time AI solution.